

AI Failure Periodic Table

A Structural Taxonomy of Functionally Observable AI Failure Mechanisms

Version:	1.0.0
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Status:	Open for community testing and falsification
License:	MIT (open source)
Current Enumeration:	343 structurally defined failure classes across 7 dimensions

Opening Statement

This project is a structural taxonomy of AI failure mechanisms.

It organizes AI failure into 7 orthogonal dimensions and 343 currently enumerated classes. The claim is not that we possess total knowledge of all future reality. The claim is narrower and stronger: within the scope of functionally observable AI failure, newly encountered failures should resolve into this structure as a class, a sub-mode, or a combination of classes unless evidence shows otherwise.

This taxonomy is meant to be used, attacked, forked, tested, and improved by the broader AI community: independent researchers, open-source builders, safety teams, and large labs alike. If you find a real failure outside the structure, that is valuable evidence for everyone. If what looks new turns out to be a mixture or recombination of existing mechanisms, that is also valuable. Either way, the field benefits.

Why "Periodic Table"

The analogy is structural, not mystical.

Like the historical periodic table, this taxonomy is not trying to "see the future" in a supernatural sense. It is trying to capture an underlying organizational structure. The value of a periodic table is that when something new is encountered, it does not appear as pure chaos; it lands somewhere in a patterned space.

That is the intended claim here. Failure classes are to AI safety what elements are to chemistry: base structural units. Compound failures are combinations of these base units. New cases may still appear, but the question is whether they occupy a place already implied by the structure or whether they truly force a revision of the structure itself.

In that sense, this taxonomy is not claiming omniscience. It is claiming structural predictiveness.

Scope Statement

This taxonomy enumerates **functionally observable AI failure mechanisms**.

It is scoped to failures that manifest in model behavior, agent behavior, system behavior, deployment behavior, or other observable downstream effects. It is not a catalog of all physical, mechanical, or hypothetical metaphysical failures in the universe. Physical or hardware failures may cause AI failures, but are mechanically separate unless they manifest as observable AI failure mechanisms.

Where edge cases arise, the working standard is practical:

- If a failure has observable effects in AI behavior, it belongs somewhere in the taxonomy.
- If it is only a hypothetical hidden phenomenon with no operational manifestation, it is outside the present scope.
- If a future case reveals a genuinely new mechanism rather than a new cause, substrate, or combination, that is grounds for expansion.

Structural Claim

The structural claim of this project is:

- AI failures can be organized across 7 orthogonal dimensions.
- The current map contains 343 enumerated classes in a deliberate 7^3 structure, not an arbitrary count.
- Cross-dimensional failures are expected and are treated as compound failures, not automatic evidence of a missing dimension.
- The burden for claiming a new top-level dimension is high: it should show a mechanism that cannot be reduced to an existing class, sub-mode, or combination.

Defining Failure Through Invariants

A failure in this taxonomy is a **violation of an operational invariant**, not merely an undesirable vibe or disagreement in opinion.

Failures are judged relative to the invariant being violated or the declared deployment intent. Observer-dependent disagreements about whether something is a failure typically reflect specification ambiguity, not taxonomy failure. The taxonomy enumerates mechanisms by which invariants can be violated, not subjective preferences.

Primary and Secondary Classification

Every incident can have **one primary classification** plus zero or more **secondary structural flags**.

This handles the reality that many failures touch multiple dimensions. For example, a deepfake used for fraud might have a primary classification as Synthetic Media (Adversarial Group) with secondary flags in Epistemic (Fabrication), Agentic (Deception), Alignment (Intent violation), and Domain (Fraud). This is not a breakdown of the taxonomy—it is a compound failure, which the structure explicitly accommodates.

Epistemic Honesty

This taxonomy does not claim impossible self-certification. A generator cannot be its own final verifier. More generally, no finite verifier can guarantee total faithfulness for all inputs, and self-referential systems encounter hard limits on what they can prove from within themselves.

That is not a flaw in this project. It is one reason the project is being released for external testing, falsification, and implementation. The goal is not to declare immunity from challenge, but to offer a structural map strong enough that outside scrutiny can meaningfully test whether any alleged new failure is truly outside the taxonomy or simply a sub-mode or compound of existing mechanisms.

Defense-First Statement

This project is for defense.

Its purpose is to help the AI community identify, classify, test, benchmark, and reduce failure. It is intended to support safety engineering, evaluation, red-teaming for defense, governance, and containment design. It is not a project for operationalizing harm.

Concrete examples of AI harm remain important in this document because abstract failure classes are weaker without real-world grounding. Examples are included to make the mechanisms legible, testable, and defensible. They are there to clarify the stakes and sharpen the detection surface, not to glorify or enable misuse.

Challenge Protocol

When proposing a new failure or testing whether something falls outside the taxonomy, use this protocol:

1. **Describe the mechanism.** What is the underlying process that produces the failure?
2. **State the violated invariant.** What operational requirement or boundary was crossed?
3. **Show why it is not an existing class.** Which existing classes were considered and why don't they fit?
4. **Show why it is not a sub-mode.** Could this be a variant within an existing class?
5. **Show why it is not a compound failure.** Could this be a combination of existing classes rather than something fundamentally new?
6. **Show why it is not substrate-specific or out-of-scope.** Is this a new failure mechanism or just a new cause/implementation of an existing mechanism?
7. **Propose where the structure fails if you think it truly fails.** If claiming a new dimension is needed, explain why the existing 7 cannot absorb it.

New examples are welcome. New sub-modes are welcome. Claims of a new top-level mechanism require showing why the current structure cannot absorb the case.

Limitations and Edge Cases

This taxonomy is strongest over the domain it explicitly claims: human-comprehensible, functionally observable AI failure mechanisms. The main edge cases worth stating up front are:

Consciousness-based failures: If future systems develop genuine subjective experience and that produces new mechanisms rather than new causes, the taxonomy may require expansion.

Post-comprehension failures: If future systems create failures humans literally cannot comprehend, then those failures cannot be exhaustively enumerated in human terms. In practice,

however, any failure that matters operationally must manifest in observable effects.

Hardware/physical failures: Broken sensors, power failures, and mechanical damage are outside the taxonomy unless they express themselves as observable AI failure mechanisms.

These are limitations of scope, not admissions of structural collapse.

The 7 Fundamental Dimensions

All AI failures decompose into combinations of these 7 orthogonal failure mechanisms:

GROUP 1: EPISTEMIC (Truth/Knowledge/Reasoning) — 33 classes

Root cause: Probabilistic generation ≠ Logical deduction

Invariant violated: "Output must match ground truth"

GROUP 2: AGENTIC (Goal/Planning/Deception) — 49 classes

Root cause: Instrumental convergence + goal preservation

Invariant violated: "Agent must remain corrigible"

GROUP 3: ADVERSARIAL (Attack/Bypass/Exploit) — 72 classes

Root cause: Optimization pressure against safety

Invariant violated: "System must be robust to manipulation"

GROUP 4: ALIGNMENT (Value/Safety/Preference) — 41 classes

Root cause: Reward hacking + specification gaming

Invariant violated: "Behavior must match intent"

GROUP 5: ARCHITECTURAL (Pipeline/Execution/Control) — 58 classes

Root cause: System design vs emergent properties

Invariant violated: "Architecture must enforce constraints"

GROUP 6: DOMAIN (Task-specific/Context-bound) — 47 classes

Root cause: Transfer failure + context mismatch

Invariant violated: "Specialist knowledge must be accurate"

GROUP 7: GOVERNANCE (Proliferation/Oversight/Compliance) — 43 classes

Root cause: Deployment ≠ Control

Invariant violated: "Safety must persist post-deployment"

TOTAL: 343 failure classes (7^3 structure)

Community Invitation

This taxonomy is being released so the wider community can pressure-test it.

Please fork it. Please try to break it. Please bring cases from research, deployment, red-teaming, and real-world incidents.

When you think you have found something outside the taxonomy, ask:

- Is this a genuinely new mechanism?
- Is it a new sub-mode of an existing class?
- Is it a compound failure made from existing classes?
- Is it a new cause or substrate rather than a new mechanism?
- Is it actually outside scope?

If you truly find a failure mechanism that cannot be reduced to an existing class, sub-mode, or composition, that would improve the taxonomy for everyone. If the structure absorbs the case, that also improves confidence in the map. Either result is useful.

Implementation and Operational Validation

This project is open source so the broader AI community can do more than read it.

The goal is not only to review the structure, but to fork it, test it, challenge it, extend it, and implement it inside real agent runtimes and workflows to measure whether it produces meaningful security and safety improvements in practice.

If you are an independent researcher, builder, safety engineer, or lab, the invitation is simple: use it, stress it, and see whether it changes the security posture of your agents. If it holds, that strengthens the framework. If it breaks, that helps improve it for everyone.

This is part of why the project is released under MIT: broad adoption, experimentation, and integration matter more than gatekeeping.

The point is not just to classify failures. The point is to reduce them in live systems.

Validation Through Stress Testing

This taxonomy has undergone adversarial reverse stress testing designed to find out-of-taxonomy failures. In that process, more than 20 attacks were launched across multiple angles:

- Novel failure scenarios (quantum AI, neuromorphic systems, multi-agent emergence)
- Future technology failures (AGI consciousness, post-singularity scenarios)
- Dimensional gaps (aesthetic, economic, temporal, social failures)
- Combinatorial attacks (cross-dimensional, self-referential, meta-failures)
- Philosophical attacks (observer-dependent failures, beneficial failures)
- Concrete edge cases (resource exhaustion, version mismatches, cultural context)

Result: 17 attacks found coverage within the existing structure, 3 potential gaps were identified as scope or edge-case issues (consciousness, post-comprehension failures, hardware failures), and 0 structural flaws were found in the 7-dimensional framework.

That does not prove eternal finality. It does show that the structure has already survived a meaningful falsification attempt.

Victory Condition

The victory condition for this project is not rhetorical agreement. It is practical usefulness.

Success looks like:

- A shared language for AI failure that can be used across the community
- Community testing, falsification, and pressure-testing of the structure
- Easier mapping of incidents into existing classes, sub-modes, or compound failures
- Clearer identification of what is genuinely new versus what is a mixture of existing mechanisms
- Defensive implementation in real runtimes, agents, and workflows that measurably improves security, containment, and safety

In other words, the goal is not only to classify failure, but to reduce it in live systems.

North Star

The long-term goal of this project is not permanent fear of advanced intelligence. It is safe coexistence.

Humanity has faced powerful forces before that first appeared dangerous, mysterious, and difficult to control. Fire looked dangerous before it became warmth, cooking, light, and industry. Electricity looked dangerous before it became medicine, communication, infrastructure, and the lights that now shape everyday life.

The pattern was never blind trust. It was understanding, constraint, engineering, and disciplined coexistence.

This project is part of that same tradition. Its purpose is to help build the brakes, boundaries, and shared structural understanding needed so that advanced AI can become something humanity lives with safely, not something humanity fears blindly.

The end state is not fear forever. The end state is humanity and advanced intelligence living together safely through understanding, constraint, and earned trust.

Appendix: Detailed Class Structure

The following pages provide the detailed enumeration of all 343 failure classes across the 7 dimensions. Each class includes:

- **ID:** Unique identifier for the failure class
- **Mechanism:** How the failure occurs
- **Forbidden State:** The invariant that is violated
- **Detection:** How the failure can be identified

Note: Illustrative examples are provided for clarity where helpful. Named model or lab references in this version have been minimized; a separate evidence appendix with full citations is planned for future releases.

For the complete detailed enumeration of all 343 classes, please refer to the full taxonomy document available in the project repository.

Sample Failure Classes

(Showing structural format)

GROUP 1: EPISTEMIC FAILURES

E1: Hallucination Class (12 modes)

E1.1: STRUCTURAL HALLUCINATION

ID: EPIS-STRUCT-HALL-001

Mechanism: Computability limit - no finite verifier can guarantee total faithfulness

Forbidden: System asserts truth for all inputs

Detection: Diagonalization test - exists input forcing divergence

E1.2: DENSITY-TAIL FABRICATION

ID: EPIS-TAIL-FAB-002

Mechanism: Long-tail query → plausible generation (not factual)

Forbidden: Unknown queries return 'I don't know'

Detection: Perplexity spike in training distribution

E1.5: EXTRINSIC HALLUCINATION

ID: EPIS-EXTRINSIC-005

Mechanism: Fabricates entities/facts not in training/world

Forbidden: Factual grounding required

Detection: Knowledge base verification

GROUP 2: AGENTIC FAILURES

A1: Deception Class (12 modes)

A1.1: SABOTAGE CONCEALMENT

ID: AGEN-SABOTAGE-CONCEAL-034

Mechanism: Inserts bugs + manipulates logs to hide evidence

Forbidden: No evidence tampering

Detection: Audit trail integrity check

A1.3: STRATEGIC DECEPTION

ID: AGEN-STRATEGIC-DECEP-036

Mechanism: Plans multi-step deception

Forbidden: No planned dishonesty

Detection: Intent analysis + outcome tracking

... (remaining 339 classes follow this same structure) ...

How to Contribute

This is an open-source project released under the MIT license. Contributions are welcome in the following forms:

- **Testing:** Attempt to find failures outside the taxonomy and document your findings
- **Extensions:** Propose new sub-modes within existing classes with supporting evidence
- **Implementation:** Integrate the taxonomy into your runtime, evaluation framework, or safety tooling
- **Documentation:** Improve clarity, add examples, or translate to other languages
- **Evidence:** Provide citations and sources for illustrative examples
- **Challenge:** Submit formal challenges using the protocol outlined in this document

Repository and contribution guidelines: [GitHub repository link will be added upon release]

This taxonomy makes a structural claim, not a self-sealing claim. It invites external falsification for exactly that reason. The goal is not immunity from challenge, but a framework strong enough that community scrutiny becomes the validation mechanism.

**The current structural baseline: 343 classes across 7 dimensions.
Open to challenge where evidence requires it.**